



**UNIVERSITÀ
DEGLI STUDI
DI TRIESTE**

Solving Chess Endgames

A Reinforcement Learning Approach

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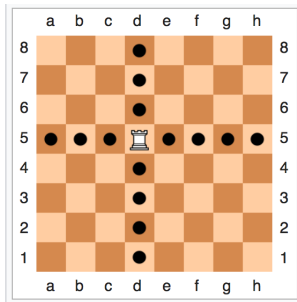
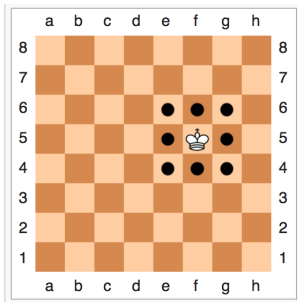
Objectives:

- Fastest checkmate against optimal player in KRk endgame
- Compare different Reinforcement Learning approaches
- Assess performance against different opponents (oracle and dummy)

Chess Programming Background

- 1890 El Ajedrecista: electro-mechanical KRK solver
- 1950 Claude Shannon: first article about chess programming
- 1997 *Deep Blue*: first computer program to beat the current world champion
- 2008 *Stockfish*: strong heuristic-based chess engine
- 2017 *Alphazero*: neural network-based approach that defeated Stockfish
- from 2018: more models developed based on *Alphazero*

Chess Rules

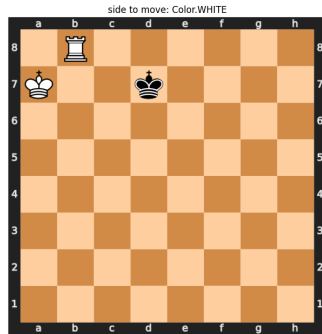


- **Win:** checkmate the opponent's king
- **Draw:** stalemate or insufficient material¹

¹threefold repetition and the fifty-move rule are not considered in our implementation

Elementary Endgames

- subset of endgames with few pieces left
- foundation for learning more complex endgames
- clear theoretical results: forced checkmate or draw
- Sygyzy endgame tablebases



Remark: white is always winning in elementary endgames!

MDP Formulation

Markov Game vs MDP

- Chess is a 2-player, turn-based, zero-sum game, formally a Markov Game
- Assume fixed black defender policy: **oracle** or **random**
- Black becomes part of the environment:
Markov Game $\rightarrow$ MDP

- State space $\mathcal{S}$: legal pieces positions and side-to-move

```
8/8/K7/8/8/5k2/7R/8 w - - 0 1
```

```
std::array<std::array<U64, 6>, 2> pieces;  
Color side_to_move;
```

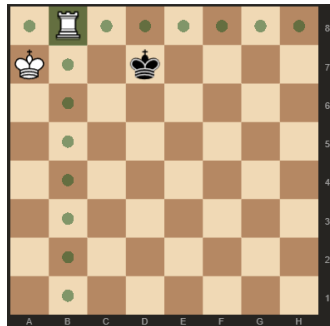
- Terminal state space $\bar{\mathcal{S}} \subset \mathcal{S}$: checkmate or draw positions (and stm). Defined procedurally

```
state.is_game_over() -> bool
```

MDP: Actions

- Action space $\mathcal{A}(s)$: set of legal moves for the current player.
- Defined procedurally:

```
state.legal_moves() ->  
std::vector<Move>
```



Transition probability $\Pr(S_{t+1} = s' | S_t = s, A_t = a)$

- **Oracle Defender:** deterministic evolution

```
state.do_move(Move)           // white deterministic
state.do_move(oracle.reply()) // black deterministic
```

- **Random Defender:** stochastic evolution

```
state.do_move(Move)           // white deterministic
state.do_move(random_reply()) // black random
```

Goal: find the fastest mate

Reward:²

- checkmate: +1
- draw: -1 (always winning for white)
- encourage faster mates
 - step penalty: -2/-0.01
 - discount factor $\gamma = 1$

²Except for Value Iteration

RL Algorithms

Value Iteration: model-based approach

- KRk: 182.676 states
- Feasible number of states + perfect knowledge $\rightarrow$ Dynamic Programming
- $V(s) \leftarrow \max_a \sum_{s',r} p(s', r|s, a)[r(s, a, s') + \gamma V(s')]$
- $\pi(s) = \arg \max_a \sum_{s',r} p(s'|s, a)[r(s, a, s') + \gamma V(s')]$

Choice of parameters

- **Step penalty:** $-2 \rightarrow$ values = DTZ
- **Draw penalty:** $-100 \rightarrow$ slow mates $>$ quick draws
- **Checkmate reward:** $+1$
- $\gamma = 1$
- $\theta = 0.01$

Q-Learning: model-free approach

- How good of a policy without knowing the “level” of the opponent?
- $Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$
- $\pi(s) = \arg \max_a Q(s, a)$
- Off-policy, online and one-step.

Choice of parameters

- **Max steps** per episode: 50 $\rightarrow$ if it's too far from mate, agent may never reach terminal state
- **Step/Draw/Mate**: -0.01/-1/+1
- $\gamma = 0.99$
- Costant **learning rate** $\alpha = 0.05 \rightarrow$ smaller variance
- ε -greedy policy with $\varepsilon = 0.3$ and $\varepsilon_{decay} = 0.99$
- $\varepsilon_t = 0.3\varepsilon_{decay}^{t+1}$

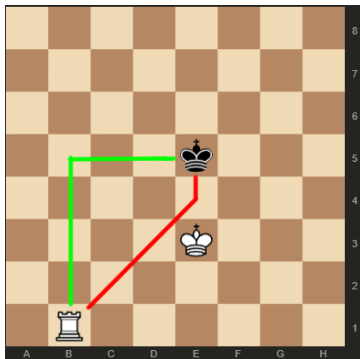
Actor-critic: approximate solution approach

- Non-elementary chess endgames → **curse of dimensionality**
- Use one-step actor-critic
- Update rules:
 - $\delta = r + \gamma \hat{V}(s'; w) - \hat{V}(s; w)$
 - $\theta \leftarrow \theta + \alpha^\theta \delta \nabla_\theta \log \pi(a|s; \theta)$
 - $w \leftarrow w + \alpha^w \delta \nabla_w \hat{V}(s; w)$

Parameters Choice

- **Policy:** MLP(128,128,64), ReLU activations, softmax output.
 - **Input:** pieces coordinates + stm (7 inputs)
- **Value function:** $\hat{V}(s; w) = \sum_{i=1}^{16} w_i \phi_i(s)$
- $\gamma = 0.99$
- **Step/Draw/Mate:** -0.01/-1/+1
- $\alpha^w = 0.02$
- $\alpha^\theta = 0.003$
- Too many iterations and no generalization $\rightarrow$ curriculum learning

Feature engineering



BK Manhattan distance
from side: maximum

BK Manhattan distance
from WR: ----

BK Chebyshev distance
from WR: ----

Opposition: True

Same rank: False

Same file: True

Monte Carlo Tree Search

- One of the most common approaches to chess
- No need to train!
- Checkmate reward: +1
- Draw penalty: -1
- Seconds per move: 10s
- Exploration constant: $c = \sqrt{2}$
- Upper confidence bound applied to trees:

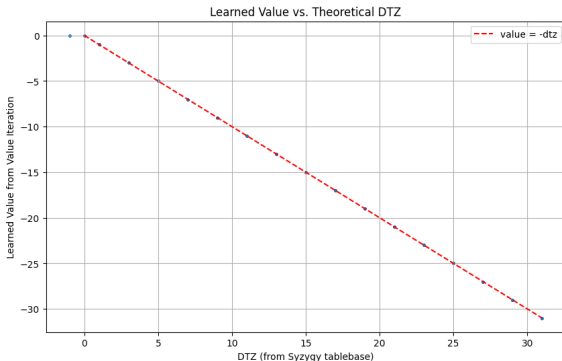
$$UCT(s, a) = Q(s, a) + c \sqrt{\frac{\ln N(s)}{N(s, a)}}$$

Results

- **Metrics:** DTZ, success rate, time per move, top-1 accuracy
- **Test set:** 10 positions per each DTZ
- **Comparison:** against **oracle** and **dummy** opponents

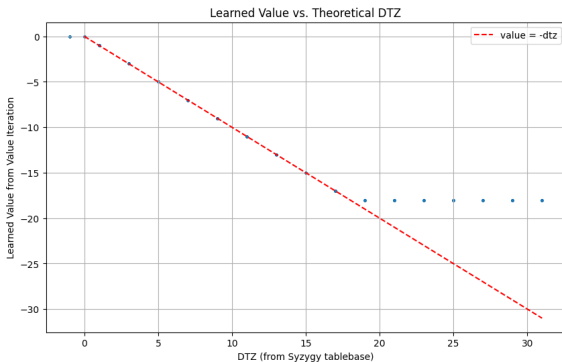
Value iterations

- Convergence to π_* after 17 iterations (~ 6 h wct on a laptop)
- Optimal policy π_* plays perfectly against an optimal opponent $\forall$ DTZ



Partial value iteration

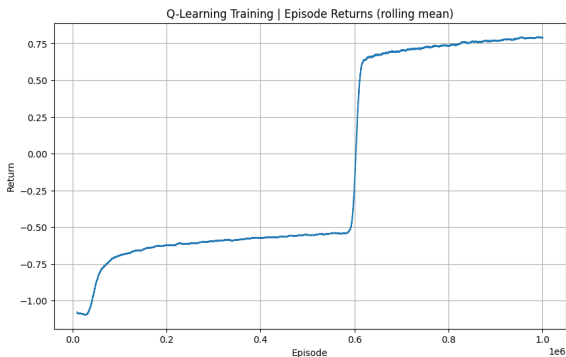
After n iterations it is able to solve mates only up to $DTZ = 2n + 1$

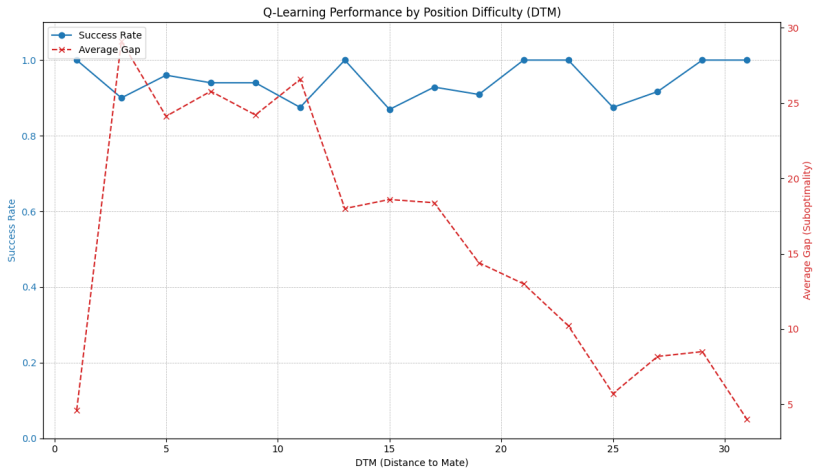


Here $n = 9$

Q-learning convergence

Convergence to **suboptimal** policy after 1M episodes
(~ 8h wct)

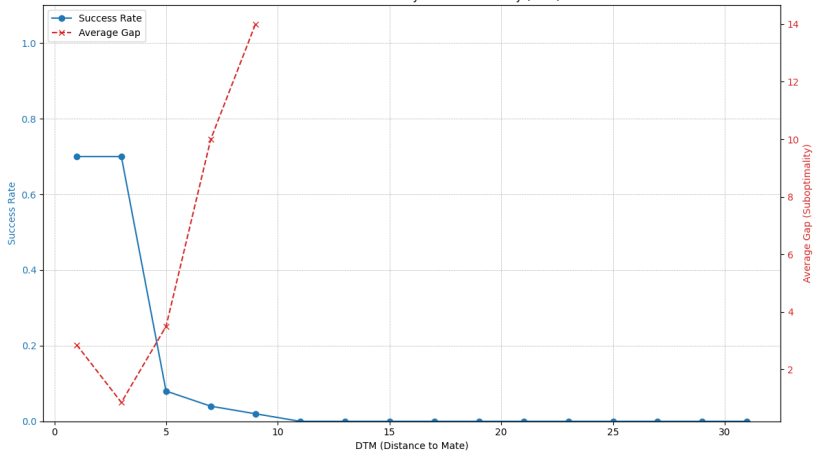




Actor-Critic convergence

- Partial convergence after 3k epochs (1.5k DTZ=1, 1.5k DTZ=3) ($\sim$ 6h wct)
- Suboptimal policy
- Explainable critic!

Actor-Critic Performance by Position Difficulty (DTM)



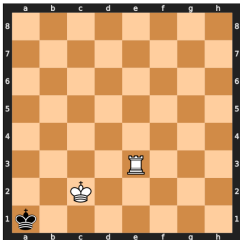
Critic through human chess principles

- $d_edgeBK = -0.269$
- $d_cornerBK = -0.402$
- $dInf(WK, BK) = -0.286$
- $dMan(WK, BK) = -0.375$
- $dInf(WR, BK) = -0.069$
- $dMan(WR, BK) = +0.200$
- $cut_w = +0.103$

With 10s of budget per move → solve mates up to DTZ=5

Top 5 white moves from root:

1.	e3a3	visits=	4897	pct=	97.9%	Q=+1.0000
2.	e3c3	visits=	10	pct=	0.2%	Q=-0.3562
3.	e3e4	visits=	8	pct=	0.2%	Q=-0.4388
4.	e3e2	visits=	5	pct=	0.1%	Q=-0.8457
5.	e3e7	visits=	5	pct=	0.1%	Q=-0.8516



Against the oracle ($DTZ = 1$)

Method	gap	success	ms_per_move	top1
VI	0.000	1.0	0.002	1.000
Q-Learning	4.600	1.0	0.002	0.823
Actor-Critic	2.857	0.7	0.865	0.639
MCTS	0.000	1.0	5155.985	1.000

Against the oracle (DTZ=3)

Method	gap	success	ms_per_move	top1
VI	0.000	1.0	0.003	1.000
Q-Learning	31.111	0.9	0.003	0.664
Actor-Critic	1.200	0.5	1.231	0.606
MCTS	0.000	1.0	7312.591	1.000

Against the Dummy (DTZ=1)

Method	gap	success	ms_per_move	top1
VI	0.000	1.0	0.004	1.000
Q-Learning	0.857	0.7	0.004	0.686
Actor-Critic	11.429	0.7	0.987	0.574
MCTS	0.000	1.0	8440.939	1.000

Against the Dummy (DTZ=3)

Method	gap	success	ms_per_move	top1
VI	0.000	1.0	0.002	1.000
Q-Learning	34.000	0.3	0.002	0.180
Actor-Critic	10.667	0.9	0.835	0.554
MCTS	2.400	1.0	8395.397	0.914

Why no convergence?

- Difficulties:
 - too many possible actions at each step
 - ideal opponent that never makes mistakes
 - very sparse positive rewards in training episodes
- Possible solutions:
 - more training time
 - implement draw by repetition and 50-move rule → more frequent terminal states

Why is it so difficult to solve chess endgames?

A player can sometimes afford the luxury of an inaccurate move, or even a definite error, in the opening or middlegame without necessarily obtaining a lost position. In the endgame ... an error can be decisive, and we are rarely presented with a second chance.

- Paul Keres

What did we learn?

- The importance of parameters choice (e.g., rewards)
- The danger of introducing a maximum length for each episode
- The importance of starting from simpler solutions and then increase complexity
- RL theory guides, but practice means dealing with gaps (e.g., hyperparameters).

- Zobrist hashing and invariance properties in chess endgames to reduce state space cardinality
- Implementing all chess rules
- Experiments with more complex chess endgames (e.g., KBBK, KRPK)
- Experiments with hyperparameters

Thank You!

Why $\gamma = 0.99$?

The longest rkr endgame requires 16 white moves, hence $\gamma^{16} \approx 0.85$ is a good reward for reaching the goal and producing fast mates.

